

Assignment 1

Assumption

1. Polytope is non-degenerate.
2. Polytope is bounded
3. Rank of A is n
4. Initial feasible point is given

Implement the simplex algorithm to maximize the objective function, You need to implement the method discussed in class.

Input: CSV file with $m+2$ rows and $n+1$ column.

The first row excluding the last element is the initial feasible point z of length n

The second row excluding the last element is the cost vector c of length n

The last column excluding the top two elements is the constraint vector b of length m

Rows third to $m+2$ and column one to n is the matrix A of size $m \times n$

Output: You need to print the sequence of vertices visited and the value of the objective function at that vertex.

- Instructions:
1. You need to implement in google Collab and share the link
 2. Implementation language is Python
 3. Input CSV file name is Testcase.csv from Collab workspace.

Assignment 2

Assumption

1. Initial feasible point is given

Implement the simplex algorithm to maximize the objective function, You need to implement the method discussed in class.

Input: CSV file with $m+2$ rows and $n+1$ column.

The first row excluding the last element is the initial feasible point z of length n

The second row excluding the last element is the cost vector c of length n

The last column excluding the top two elements is the constraint vector b of length m

Rows third to $m+2$ and column one to n is the matrix A of size $m \times n$

Output: You need to print the sequence of vertices visited and the value of the objective function at that vertex.

- Instructions:
1. You need to implement in google Collab and share the link
 2. Implementation language is Python
 3. Input CSV file name is Testcase.csv from Collab workspace.

Assignment 3

Implement the simplex algorithm to maximize the objective function, You need to implement the method discussed in class.

Input: CSV file with $m+1$ rows and $n+1$ column.

The first row excluding the last element is the cost vector c of length n

The last column excluding the top element is the constraint vector b of length m

Second to $m+1$ rows and column one to n is the matrix A of size $m \times n$

Output: You need to print the initial feasible point, sequence of vertices visited and the value of the objective function at that vertex.

- Instructions:
1. You need to implement in google Collab and share the link
 2. Implementation language is Python
 3. Input CSV file name is Testcase.csv from Collab workspace.

Assignment 4

Implement algorithm to find minimum weight perfect matching in a edge weighted bipartite graph, you need to implement the method discussed in class.

Input: CSV file with 3 rows and number of columns is number of edges.

Each column corresponds to a edge.

First and second rows are vertex ids of the edges, and third row is the edge weights.

Note that vertex ids are integers.

Output: You need to print values of dual variables at each iteration.

- Instructions:
1. You need to implement in google Collab and share the link
 2. Max two members in a group. You need to mention roll number of group members at the top of the code

Assignment 5

Implement algorithm to find minimum cost max flow in a network. You need to implement the method discussed in class.

Input: CSV file with 4 rows and number of columns is number of edges.

Each column corresponds to a edge.

First row is the 'from' vertex and second row is the 'to' vertex of edges, and third row is the capacity of the edges, and fourth row is the cost of sending one unit flow on that.

Note that vertex ids are integers.

Source vertex id is '1' and destination vertex id is '2'

Output: You need to print total cost at each iteration.

Instructions: 1. You need to implement in google Collab and share the link

2. Max two members in a group. You need to mention roll number of group members at the top of the code